7.1 Activities: numbers to a million

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Pick one activity per session. None of them is timed. Let him be the "teacher" sometimes: explaining something to you is one of the best ways to learn it.

1. Biggest number (cards)

You need: a deck of cards with the picture cards removed (ace = 1, ten = 0), and a sheet with six boxes labelled HD | TD | D | H | T | E.

How: Each player turns over cards one at a time and must place each card in a box *before* seeing the next one. Once placed, a card can't move. After six cards, read your number out loud. Biggest number wins the round.

Variations: smallest number wins; closest to 500 000 wins; a "secret target" chosen by him.

Talk about it: "You turned over a 9. Why did you put it there?" "Was it a good move or bad luck?" This is the same game as "Biggest number" in GAMES.md, stretched to six digits.

2. Big numbers hunt

You need: a newspaper, a house-for-sale website, the car's odometer, a train or bike-route map, the town website.

How: Hunt for numbers over 10 000. Write them on sticky notes. Read each one out loud together and sort them from smallest to largest on the fridge. Good finds: the price of a house in your street (often hundreds of thousands of euros), your town's population, the kilometres on the car's odometer, a football crowd.

Talk about it:

- "Is this number exact, or has someone rounded it? How can you tell?"
- "Would a newspaper say 54 865 people or 'about 55 000'? Why?"

3. Washing-line number line

You need: a string or washing line across the room, pegs, cards.

How: Peg 0 at one end and 1 000 000 at the other. Ask him where 500 000 goes. Then 100 000, 250 000, 900 000. Then give him cards with numbers like 386 000, 47 000, 999 999, and let him peg them. Change the ends: 300 000 to 400 000, so 386 000 now goes near the right.

Talk about it: "How did you decide where to put it? What did you use to help?" (Usually: halfway first.)

4. The ×10 machine (calculator)

You need: a calculator or phone calculator, paper.

How: He writes a number. Before pressing anything, he predicts what × 10 will give. Then he checks on the calculator. Do the same for × 100, × 1000, : 10, : 100. Put the results in a place-value table and look at how the digits move.

Talk about it: "What is the rule? Does it always work?" Then try 2,5 × 10 together. "Did 'add a zero' work here? What *does* always work?" (The digits move one place.)

5. Guess my number

You need: nothing, or paper for a number line.

How: You think of a number between 0 and 1 000 000. He asks yes/no questions or guesses and you say “higher” or “lower”. He can draw a number line and cross out parts. Swap roles.

Talk about it: “What was your best first guess? Why is 500 000 a clever start?” Count how many guesses it took, but as a fun fact, not a score.

6. A million rice grains?

You need: a 1 kg bag of rice, kitchen scales, a small bowl.

How: Weigh out 10 g of rice. Count the grains together (make piles of 10, then piles of 100). Then work out roughly how many grains are in the whole kilogram (1 kg is 100 times 10 g). Is it more or less than a million? How many bags would you need for a million?

Talk about it: “We counted roughly. Is that OK here? When does a number need to be exact?”

Conversation prompts for any day

- “What’s a number you saw today that was more than a thousand?”
- “Is a million seconds more or less than a week? Guess first.” (See the puzzle corner in practice.md.)
- “When would you round a number, and when wouldn’t you?” (Prices on a receipt: exact. Crowd at a match: rounded.)